

Quotients in Algebraic Geometry

From categorical quotients to schemes, algebraic spaces, and quotient stacks

PRESENTED BY

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*Throughout this presentation, k denotes an algebraically closed field.
(Yes, k is always as nice as we want it to be.)*

What Should X/G Mean?

Let S be a scheme, G an S -group scheme, and X an S -scheme equipped with an action

$$a : G \times_S X \longrightarrow X.$$

The quotient problem

We want to construct a geometric object Q and a morphism

$$q : X \longrightarrow Q$$

which identifies points belonging to the same G -orbit.

The morphism q must be G -invariant:

$$q \circ a = q \circ \text{pr}_2.$$

The central questions

- What universal property should characterize Q ?
- Does Q exist in the category of schemes?
- If not, which larger category should contain it?

Categorical Quotient

Let $\mathcal{C}$ be a category and let a group (ordinary) G act on an object $X \in \mathcal{C}$.
A morphism $q : X \rightarrow Q$ is called G -invariant if

$$q \circ g = q \quad \text{for every } g \in G.$$

Definition

A G -invariant morphism

$$q : X \longrightarrow Q$$

is called a **categorical quotient** if, for every object $Y \in \mathcal{C}$ and every G -invariant morphism $f : X \rightarrow Y$, there exists a unique morphism

$$\bar{f} : Q \longrightarrow Y$$

such that

$$f = \bar{f} \circ q.$$

Equivalently, composition with q induces a natural bijection

$$\mathrm{Hom}_{\mathcal{C}}(Q, Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(X, Y)^G.$$

Quotients in Sets and Topological Spaces

Quotient of Sets

Let a group G act on a set X . The quotient is the set of G -orbits:

$$X/G = \{Gx \mid x \in X\}, \quad q : X \longrightarrow X/G, \quad q(x) = Gx.$$

Thus, two points have the same image precisely when they belong to the same orbit:

$$q(x) = q(y) \iff y = gx \text{ for some } g \in G.$$

Quotient of Topological Spaces

Suppose X is a topological space and the action of G is continuous. We equip the orbit set X/G with the **quotient topology**:

$$U \subseteq X/G \text{ is open} \iff q^{-1}(U) \subseteq X \text{ is open.}$$

This is the finest topology for which the projection $q : X \rightarrow X/G$ is continuous.

Quotients of Affine Schemes

Let $X = \operatorname{Spec}(A)$ be an affine scheme, and suppose that a group G acts on X by scheme automorphisms. Since Spec is contravariant, this induces an action of G on the ring A .

Construction

The quotient in the category of affine schemes is

$$X/G := \operatorname{Spec}(A^G).$$

The inclusion of rings

$$A^G \hookrightarrow A$$

induces the canonical G -invariant morphism

$$q : \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(A^G).$$

For every affine scheme Y , composition with q induces

$$\operatorname{Hom}(X/G, Y) \cong \operatorname{Hom}(X, Y)^G.$$

Thus, $\operatorname{Spec}(A^G)$ is the categorical quotient in $\mathbf{AffSch}$.

Quotients of Ringed Spaces

Let G be a group acting on a ringed space $(X, \mathcal{O}_X)$ by automorphisms. First form the topological orbit space

$$q : X \longrightarrow Y := X/G$$

with the quotient topology.

Construction of the structure sheaf

For every open subset $U \subseteq Y$, define

$$\mathcal{O}_Y = (q_* \mathcal{O}_X)^G,$$

Equivalently, $\forall U \in \text{Ouv}(Y) : \mathcal{O}_Y(U) := \mathcal{O}_X(q^{-1}(U))^G$.

The inclusion $\mathcal{O}_Y \hookrightarrow q_* \mathcal{O}_X$, makes $q : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ a morphism of ringed spaces.

Quotients of Ringed Spaces

Local structure

If $(X, \mathcal{O}_X)$ is a locally ringed space, then $(Y, (q_*\mathcal{O}_X)^G)$ is also a locally ringed space.

Proposition

Let G be a finite group acting on a ring A by ring automorphisms. Set $X = \operatorname{Spec}(A)$, $B = A^G$, and let $p : X \rightarrow \operatorname{Spec}(B)$ be the morphism induced by the inclusion $B \hookrightarrow A$.

Then p induces a homeomorphism $X/G \xrightarrow{\sim} \operatorname{Spec}(A^G)$

Moreover, the canonical morphism of sheaves $\mathcal{O}_{\operatorname{Spec}(A^G)} \rightarrow (p_*\mathcal{O}_X)^G$ is an isomorphism.

Consequently, the ringed-space quotient

$$(X/G, (q_*\mathcal{O}_X)^G)$$

is a scheme, and there is a canonical isomorphism

$$X/G \cong \operatorname{Spec}(A^G).$$

Finite Group Actions and Quotients

Proposition

Let X be a scheme with an action of a finite group G , and let $p : X \longrightarrow Y$ be a G -invariant affine morphism such that $\mathcal{O}_Y \xrightarrow{\sim} (p_* \mathcal{O}_X)^G$. Then:

- (i) p is an **integral** morphism;
- (ii) p is **surjective**, its fibres are exactly the G -orbits, and the topology of Y is the quotient topology of X ;
- (ii) (Y, p) is a **quotient scheme** of X by G .

Definition

Let a finite group G act on a scheme X . The action is called **admissible** if there exists a morphism $p : X \longrightarrow Y$ satisfying the hypotheses of Proposition above. In particular, X/G exists and $X/G \simeq Y$.

Criterion for an Admissible Action

proposition

Let a finite group G act on a scheme X . Then the action is admissible if and only if

$$X = \bigcup_{\alpha} U_{\alpha},$$

where each $U_{\alpha} \subseteq X$ is an **G -invariant affine open subscheme**.

Equivalently, every G -orbit in X is contained in an affine open subscheme.

Exercise 3.6(1)

Let $X = \operatorname{Spec}(A)$, and let $U \subseteq X$ be an open subscheme stable under the action of G . The quotient scheme U/G exists and that

$$U/G \simeq p(U).$$

Geometric Quotients

Definition (Geometric Quotient)

Let a group G act on a scheme X . A quotient morphism

$$\pi : X \longrightarrow Y$$

is called a **geometric quotient** if:

- 1 π is a **categorical quotient**;
- 2 the fibres of π are exactly the G -orbits;
- 3 the topology of Y is the quotient topology of X ;
- 4 regular functions on Y correspond to G -invariant regular functions on X .

Exercise 3.8

If the action of G has no fixed points, then the quotient is **geometric**; in particular,

$$\pi^{-1}(\pi(x)) = G \cdot x \quad \text{for every } x \in X.$$

Failure of Quotients: Invariant Functions Lose Information

Example: Translation Action

Consider the action $\mathbb{Z} \curvearrowright \mathbb{A}_k^1$ given by $n \cdot x = x + n$. A categorical quotient in affine schemes is obtained from the invariant ring: $k[x]^{\mathbb{Z}}$.

If $f(x+1) = f(x)$, then (in characteristic 0) f must be constant.

Therefore: $k[x]^{\mathbb{Z}} = k$. Hence the affine categorical quotient is $\mathbb{A}^1/\mathbb{Z} = \text{Spec}(k)$, a single point.

Failure

The quotient obtained from invariant functions does not reflect the orbit space.

The action identifies points differing by integers: $x \sim x + n$,

but all this information disappears in $\text{Spec}(k)$.

Therefore invariant functions may fail to separate different orbits.

Categorical quotient may be too small geometrically

Failure of Geometric Quotients: Orbit Separation

Example: $\mathbb{G}_m$ acting on $\mathbb{A}^2$

Consider the action $t(x, y) = (tx, t^{-1}y)$.

The invariant ring is

$$k[x, y]^{\mathbb{G}_m} = k[xy],$$

hence the categorical quotient is

$$q : \mathbb{A}^2 \longrightarrow \mathbb{A}^1, \quad q(x, y) = xy.$$

Failure

For $c \neq 0$, the fibre $q^{-1}(c)$ is a single orbit. However, $q^{-1}(0) = \{xy = 0\}$ is the union of the two coordinate axes, containing different $\mathbb{G}_m$ -orbits.

Therefore:

fibres of the categorical quotient are not always orbits

Hence the quotient is not a geometric quotient.

Failure of Schemes to Remember Symmetries

Example: $\mathbb{A}^2/\{\pm 1\}$

Let $G = \mathbb{Z}/2\mathbb{Z}$ act on $\mathbb{A}^2 = \operatorname{Spec} k[x, y]$ by $(x, y) \mapsto (-x, -y)$.

The invariant ring is

$$k[x, y]^G = k[x^2, xy, y^2].$$

Hence the quotient is

$$X/G = \operatorname{Spec} \frac{k[u, v, w]}{(uw - v^2)}.$$

Failure

The original space $\mathbb{A}^2$ is smooth, but the quotient has a singularity at $(0, 0, 0)$. Moreover the quotient remembers the singular point but forgets the symmetry responsible for it:

$$\operatorname{Stab}(0, 0) = \mathbb{Z}/2\mathbb{Z}.$$

Quotient schemes lose stabilizer information

This motivates algebraic spaces and quotient stacks.

The Quotient Functor

Motivation

The quotient X/G as a set only remembers the points after identification. However, a scheme is not determined only by its points. It is determined by how it behaves with respect to all test schemes.

For a scheme T , consider the functor of points:

$$T \longmapsto X(T) = \operatorname{Hom}(T, X).$$

A more faithful quotient should describe families of orbits:

$$T \longmapsto X(T)/G(T)$$

rather than only the set of orbits: X/G .

The problem

Does there exist a scheme Y such that $\operatorname{Hom}(T, Y) \simeq X(T)/G(T)$ for every scheme T ?

Algebraic Spaces: Quotients Beyond Schemes

The problem

Sometimes the quotient construction is natural, but the resulting object is not a scheme. In other words:

The quotient exists, but not inside **Sch**.

Definition (Idea of Algebraic Spaces)

An algebraic space is a geometric object which is locally described by quotients of schemes. It extends the category of schemes and allows quotient constructions.

Why do we need them?

Many quotients of the form X/G especially for well-behaved actions (for example free actions), may exist naturally as algebraic spaces but not as schemes.

Algebraic spaces solve the problem: “the quotient exists, but not as a scheme.”

Quotient Stacks: Remembering Symmetry

The remaining problem

Even algebraic spaces may forget important information.

A scheme quotient remembers:

points after identification

but it forgets:

automorphisms and stabilizers.

Definition (Quotient Stack)

For a group action $G \curvearrowright X$, the quotient stack $[X/G]$ is designed to remember both:

- the orbit of each point;
- the symmetry group acting on that point.

Quotient Stacks: Remembering Symmetry

Main idea

A point of $[X/G]$ is not only an orbit.

It is an orbit together with its automorphism group.

Example: remembering stabilizers

Consider the action $\mathbb{Z}/2\mathbb{Z} \curvearrowright \mathbb{A}^2$, $(x, y) \mapsto (-x, -y)$.

In the ordinary quotient, $(0, 0)$ becomes only a singular point.

The quotient stack remembers that

$$\mathrm{Stab}(0, 0) = \mathbb{Z}/2\mathbb{Z}.$$

$$[X/G] = \text{points} + \text{functions} + \text{symmetries}$$

Final Perspective: What is a Quotient in Algebraic Geometry?

Naive idea

Given a group action $G \curvearrowright X$, we identify points in the same orbit: $x \sim g \cdot x$.

The problem

In algebraic geometry, a quotient should preserve more than points:

- geometric structure (functions);
- behaviour under families of points;
- symmetries and stabilizers.

Quotient notion	Information preserved
Orbit space	Orbits of points
Categorical quotient	Universal property + invariant functions
Geometric quotient	Categorical quotient + fibres = orbits
Algebraic space	Quotients beyond schemes
Quotient stack $[X/G]$	Orbits + symmetries